

**ARCHITECTURE
DOCUMENT
STELOS**

1. INTRODUCTION

1.1 Purpose

STELOS (Steel Operations Intelligence System) is an Agentic AI-powered maintenance intelligence platform developed to support industrial equipment maintenance in steel manufacturing environments. The system combines Large Language Models (LLMs), multi-agent orchestration, predictive analytics, anomaly detection, and Retrieval-Augmented Generation (RAG) to assist maintenance engineers in making faster, more accurate, and explainable maintenance decisions.

The primary objective of STELOS is to reduce unplanned downtime, improve maintenance planning, enhance diagnostic accuracy, and enable a transition from reactive maintenance to proactive and predictive maintenance strategies. By continuously analyzing equipment telemetry, historical maintenance records, technical manuals, and operational constraints, the platform provides intelligent recommendations for fault diagnosis, root cause analysis, risk assessment, Remaining Useful Life (RUL) prediction, and maintenance planning.

1.2 Scope

The scope of STELOS covers critical industrial assets commonly found in steel manufacturing plants, including blast furnaces, rolling mills, pumps, conveyors, compressors, cooling systems, and auxiliary equipment.

The platform supports:

- Real-time monitoring of equipment health
- Detection of abnormal operating conditions
- Root cause analysis of failures
- Remaining Useful Life estimation
- Risk and criticality assessment
- Maintenance planning and optimization
- Spare parts procurement recommendations
- AI-powered conversational troubleshooting
- Executive-level maintenance reporting

The system is designed as a decision-support platform and does not directly control physical equipment.

2. ARCHITECTURAL GOALS AND CONSTRAINTS

2.1 Reliability

Industrial maintenance systems must operate continuously and provide accurate recommendations even under varying operational conditions. STELOS is designed to maintain high reliability through modular agent orchestration, fault-tolerant workflows, and explainable reasoning mechanisms. The system prioritizes consistency and traceability of recommendations to ensure trust among maintenance engineers.

2.2 Scalability

Steel manufacturing plants often contain hundreds of monitored assets generating large volumes of sensor data. The architecture is designed to scale horizontally, allowing multiple assets, users, and concurrent maintenance investigations to be processed simultaneously. The modular multi-agent architecture ensures that additional agents and analytical capabilities can be integrated without impacting existing workflows.

2.3 Explainability

Maintenance decisions must be transparent and auditable. Every recommendation generated by STELOS includes confidence scores, evidence chains, retrieved documentation references, and contextual explanations. This allows engineers to understand why a recommendation was generated and supports informed decision-making.

3. PRODUCTIZATION ASSESSMENT

3.1 Reusable Components

The platform is built using reusable and modular components that can be deployed across multiple industrial environments. Components such as the RAG pipeline, anomaly detection engine, RUL prediction module, agent orchestration framework, and reporting services are independent and reusable across various maintenance use cases.

3.2 Open Source Technologies Used

STELOS leverages modern open-source technologies including FastAPI, React, TypeScript, LangGraph, LangChain, FAISS, Scikit-Learn, SQLite, and Docker. These technologies provide scalability, maintainability, and deployment flexibility while reducing implementation costs.

3.3 LLM and Agent Framework Assessment

The platform uses Gemini as the reasoning engine and LangGraph as the orchestration framework. Gemini provides contextual understanding, maintenance reasoning, and report generation capabilities, while LangGraph enables structured multi-agent collaboration across diagnostic, planning, procurement, and reporting workflows.

4. SYSTEM ARCHITECTURE

4.1 Overview

STELOS follows a layered architecture consisting of User Interface Services, Agentic AI Services, Predictive Analytics Services, Knowledge Retrieval Services, and Data Services.

At the user layer, maintenance engineers interact through a web-based dashboard that provides operational visibility, conversational troubleshooting, and maintenance planning capabilities.

The backend layer hosts the Agentic AI framework, which orchestrates multiple specialized agents responsible for diagnostics, risk analysis, planning, procurement intelligence, and reporting.

The analytics layer performs anomaly detection, Remaining Useful Life prediction, and risk scoring using machine learning models.

The knowledge layer utilizes Retrieval-Augmented Generation (RAG) to retrieve relevant information from maintenance manuals, SOPs, historical logs, and failure reports.

The data layer stores telemetry, maintenance history, agent feedback, and knowledge-base content required for reasoning and decision support.

5. AGENTIC AI DESIGN

5.1 Supervisor Agent

The Supervisor Agent serves as the central orchestration component of STELOS. It receives user queries, equipment alerts, and operational events, determines the required workflow, and coordinates interactions between specialized agents. The Supervisor Agent ensures efficient execution of maintenance investigations while maintaining context throughout the decision-making process.

5.2 Diagnostic Agent

The Diagnostic Agent analyzes sensor telemetry, operational logs, and anomaly indicators to identify potential equipment faults. It evaluates abnormal operating conditions and generates preliminary diagnostic assessments supported by confidence scores and supporting evidence.

5.3 Knowledge Retrieval Agent

The Knowledge Retrieval Agent utilizes Retrieval-Augmented Generation (RAG) to access maintenance manuals, SOPs, failure reports, and historical maintenance records. It provides contextual knowledge required for accurate maintenance recommendations.

5.4 Root Cause Analysis Agent

The Root Cause Analysis Agent correlates diagnostic findings, historical incidents, and retrieved documentation to identify the most probable causes of equipment failures. The agent provides explainable reasoning and supporting evidence for each identified root cause.

5.5 RUL Prediction Agent

The Remaining Useful Life (RUL) Prediction Agent estimates the operational lifespan of equipment based on degradation patterns, maintenance history, and predictive analytics models.

5.6 Risk Assessment Agent

The Risk Assessment Agent evaluates operational, financial, and safety risks associated with identified equipment issues. It prioritizes maintenance actions based on criticality and business impact.

5.7 Maintenance Planning Agent

The Maintenance Planning Agent generates immediate corrective actions, preventive maintenance schedules, and long-term monitoring strategies.

5.8 Procurement Intelligence Agent

The Procurement Intelligence Agent evaluates spare-part availability, procurement lead times, and inventory constraints to support maintenance planning.

5.9 Reporting Agent

The Reporting Agent consolidates outputs from all agents and generates structured maintenance reports, executive summaries, and decision-support documents.

6. DATA FLOW AND SYSTEM FLOW

6.1 Sensor Data Ingestion

Equipment telemetry data is continuously collected from operational systems and condition monitoring sources. This data includes temperature, vibration, pressure, current consumption, and other health indicators.

6.2 Anomaly Detection Pipeline

Incoming telemetry is processed through anomaly detection models to identify abnormal operating patterns. Potential anomalies are forwarded to the agent orchestration layer for investigation.

6.3 Multi-Agent Orchestration Flow

The Supervisor Agent coordinates task execution across specialized agents. Information is shared through a structured workflow that ensures contextual consistency.

6.4 RAG Retrieval Pipeline

Relevant maintenance knowledge is retrieved from indexed documents and supplied to reasoning agents as contextual evidence.

6.5 Maintenance Recommendation Flow

Agent outputs are consolidated to generate maintenance recommendations and operational guidance.

6.6 Report Generation Flow

Final recommendations, risk assessments, and maintenance plans are converted into structured reports for engineers and supervisors.

7. MODEL DESIGN AND REASONING PIPELINE

7.1 LangGraph Orchestration

LangGraph provides workflow orchestration for the multi-agent system. It enables state management, agent routing, and context preservation.

7.2 LLM Reasoning Layer

The LLM layer performs contextual reasoning, maintenance planning, explanation generation, and conversational interactions.

7.3 Isolation Forest Anomaly Detection

Isolation Forest models identify unusual equipment behavior and detect early signs of degradation.

7.4 Weibull RUL Prediction

Weibull reliability analysis estimates Remaining Useful Life and equipment degradation trends.

7.5 XGBoost Risk Scoring

XGBoost models estimate failure probability and business impact using operational and maintenance features.

7.6 Explainability Framework

The explainability framework generates evidence chains, source citations, and confidence indicators for every recommendation.

7.7 Confidence Score Generation

Confidence scores are computed using model outputs, retrieval quality, and agent agreement levels.

8. KNOWLEDGE RETRIEVAL (RAG) ARCHITECTURE

8.1 Knowledge Sources

Knowledge sources include equipment manuals, SOPs, maintenance logs, failure reports, and engineering documentation.

8.2 Document Processing

Documents are parsed, cleaned, segmented, and transformed into searchable chunks.

8.3 Embedding Generation

Semantic embeddings are generated for efficient retrieval of maintenance knowledge.

8.4 FAISS Vector Database

FAISS stores vectorized document representations and enables fast similarity search.

8.5 Context Retrieval

Relevant maintenance information is retrieved dynamically during agent execution.

8.6 Citation Generation

Retrieved information is linked to source references for traceability and explainability.

9. ALERTING AND PREDICTION LOGIC

The alerting framework combines anomaly detection, predictive analytics, equipment criticality, and agent reasoning to identify emerging maintenance risks.

The prediction framework continuously evaluates equipment degradation patterns and generates proactive maintenance recommendations before failures occur.

The system prioritizes alerts based on operational impact, safety considerations, maintenance urgency, and procurement constraints.

10. GENERAL TECHNICAL SERVICES

The platform incorporates persistence, real-time communication, logging, monitoring, error handling, configuration management, and security controls required for industrial-grade operation.

These services ensure reliability, maintainability, and operational visibility across the entire platform.

11. USER INTERFACE MODULES

The user interface consists of specialized operational dashboards that support maintenance investigations, asset monitoring, risk analysis, procurement planning, and executive reporting.

Each interface is designed to provide role-specific visibility while maintaining a consistent user experience.

12. DEPLOYMENT ARCHITECTURE

STELOS supports cloud-native deployment using containerized services. Frontend and backend services can be independently scaled based on workload requirements.

The deployment architecture supports secure API communication, centralized configuration management, and automated deployment workflows.

13. INSTALLATION AND CONFIGURATION GUIDE

The installation process includes backend dependency setup, frontend configuration, environment variable management, database initialization, and deployment procedures.

Comprehensive setup instructions ensure reproducibility across development, testing, and production environments.

14. SAMPLE INPUT AND OUTPUT DEMONSTRATION

Sample scenarios demonstrate how equipment telemetry and engineer queries are transformed into actionable maintenance insights.

Outputs include fault diagnosis, root cause analysis, RUL estimates, risk assessments, maintenance plans, procurement recommendations, and executive summaries.

15. ASSUMPTIONS

The platform assumes availability of operational telemetry, maintenance records, equipment documentation, and sufficient historical data for predictive analysis.

It is also assumed that maintenance personnel provide feedback to support continuous improvement.

16. RISKS AND LIMITATIONS

Model performance depends on data quality, document completeness, and operational consistency.

Certain predictions may be impacted by limited historical failure data or rapidly changing operating conditions.

Future iterations can further improve model accuracy through additional training data and operational feedback.

17. ALTERNATIVE SOLUTIONS CONSIDERED

Alternative approaches evaluated include traditional predictive maintenance systems, statistical reliability models, standalone LLM assistants, and basic RAG-based maintenance tools.

A multi-agent architecture was selected due to its superior ability to perform complex reasoning, contextual decision-making, and coordinated maintenance planning.

18. BUSINESS IMPACT

STELOS is designed to reduce unplanned downtime, improve maintenance efficiency, optimize spare-part utilization, and enhance asset reliability.

The platform supports data-driven maintenance decisions that can significantly reduce operational costs and improve plant productivity.

19. CONCLUSION

STELOS demonstrates the practical application of Agentic AI in industrial maintenance environments. By integrating predictive analytics, multi-agent reasoning, explainable AI, and knowledge retrieval, the platform enables intelligent maintenance decision support at enterprise scale.

The solution aligns with modern Industry 4.0 objectives and supports the transition from reactive maintenance practices to proactive and predictive maintenance operations.